

A

Fixed spectrum; different interactions

All 105 perfect matchings

$$f_M = \frac{1}{2} \sum_{(i,j) \in M} x_i x_j$$

All 35 four-plus-four partitions

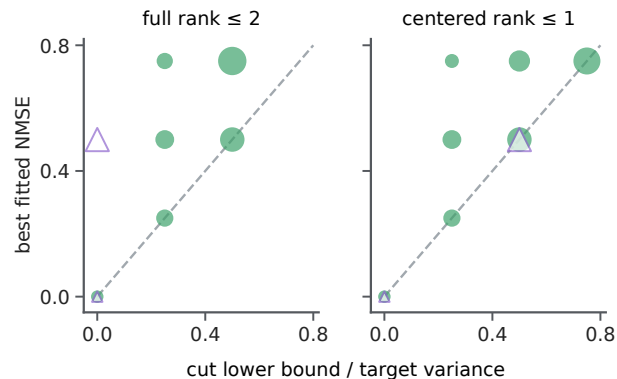
$$f_A = \frac{1}{2} \left(\prod_{i \in A} x_i + \prod_{i \notin A} x_i \right)$$

$$\mathbb{E}[\nabla_x \nabla_x^T f] = I_8/4; \text{rank} = 8$$

28 coefficients · 14 edges · root-only output

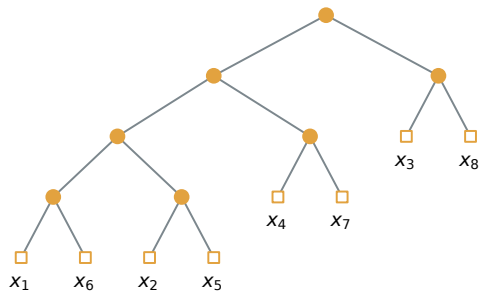
B

The centered constraint is stronger

**C**

Nested: minimum depth 4

root readout

**D**

Depth constraint over all labeled trees

